

FRANCES (SU) LIU

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RESEARCH INTERESTS

My research asks when and how different information sets sharpen volatility forecasts, combining high-frequency market microstructure data and media sentiment within a common HAR-based framework. Building on this and on six years of prior industry experience in equity strategy and portfolio management, I am also interested in extending these methods to high-dimensional portfolio allocation, combining dependence modeling with machine learning.

Core areas: financial econometrics; volatility modeling and forecasting; high-frequency and tick-level data; extended trading hours and information transmission; machine learning and deep sequence models in asset pricing; media sentiment and state-dependent predictability; portfolio allocation and dependence modeling.

EDUCATION

Doctor of Philosophy (PhD) in Finance

2024.02–2027.08 (*expected*)

University of Technology Sydney (UTS), Sydney, Australia

Supervisors: Assoc. Prof. [Vitali Alexeev](#) (UTS, Finance) & Assoc. Prof. [Katja Ignatieva](#) (UNSW, Actuarial)

Master of Science in Financial Engineering

2015.08–2017.06

Stevens Institute of Technology, New Jersey, United States

GPA: 3.9/4.0

Thesis: Systemic Risk Measurement: CoVaR and Copula Models (ARMA/TGARCH conditional moments, copula-based dependence between financial sectors, time-series CoVaR). Supervised by Assoc. Prof. [Zhenyu Cui](#) (School of Systems & Enterprises, Hanlon Lab)

Bachelor of Economics in Financial Engineering

2011.09–2015.06

Harbin University of Commerce, Harbin, China

WORKING PAPERS

Reading Between the Trades: Neural Encoding Extended-Hours Tick Sequences for Volatility Forecasting

with Vitali Alexeev (UTS), [Adam Clements](#) (QUT), and Katja Ignatieva (UNSW)

working paper

- Replaces the coarse overnight aggregates used in the volatility literature with the complete tick-level record of each extended-hours session, compressing variable-length, irregularly spaced transaction sequences into fixed-dimensional representations through neural encoders and integrating them with the HAR model
- An incremental architecture hierarchy isolates the contribution of each modeling ingredient; applied to all DJIA constituents over 2002–2025, the simplest architecture reduces out-of-sample forecast loss by approximately 11% at the one-day horizon, roughly twice the gain of the best scalar overnight proxy
- Pre-open and post-close sessions deliver comparable aggregate gains through opposite channels: 64% of the pre-open improvement is recoverable from a single within-session aggregate, while 81% of the post-close improvement is attributable to tick-sequence structure. Gains are stable across the Global Financial Crisis and COVID-19 episodes
- *JEL:* C45, C53, G12, G14

When Does Sentiment Predict Volatility?

with Vitali Alexeev (UTS) and Katja Ignatieva (UNSW)

under review, Journal of Applied Econometrics

- Augmenting HAR models with Thomson Reuters MarketPsych sentiment yields only modest average improvements, but this average masks strong state dependence
- Sentiment becomes materially more informative in the upper tail of the volatility distribution and, more importantly, when volatility persistence weakens; persistence is measured within a fractional integration framework
- Out-of-sample gains concentrate in regimes jointly characterised by high volatility and low persistence, indicating that sentiment supplies forward-looking narrative information precisely when price-based models are least reliable

- *JEL*: C22, C53, G12, G14

CONFERENCE PRESENTATIONS

When Does Sentiment Predict Volatility?

- Econometric Society Australasia Meeting (ESAM), Adelaide, Australia, November 2026 (*accepted*)
- FIRN Annual Conference, K'gari, Australia, November 2026 (*accepted*)
- International Association for Applied Econometrics (IAAE) Annual Conference, Carcavelos, Portugal, June 2026

Reading Between the Trades

- Econometric Society Australasia Meeting (ESAM), Adelaide, Australia, November 2026 (*accepted*)

AWARDS AND SCHOLARSHIPS

UTS Business Doctoral Scholarship (full stipend)	2024–2027
UTS International Research Scholarship (full tuition)	2024–2027
Academic Excellence Award, Stevens Institute of Technology	2017

TEACHING EXPERIENCE

University of Technology Sydney

Sydney, Australia

Tutor, Finance Discipline Group

2024–present

- 25503 *Investment Analysis* (undergraduate), Spring 2026
- 25556 *The Financial System* (undergraduate), Autumn 2026 and Spring 2026

INDUSTRY EXPERIENCE

Six years of applied quantitative research, carrying models from hypothesis through backtesting to deployment.

Webull Financial LLC

Changsha, China

Senior Quantitative Analyst

2020.04–2023.11

- Built factor models for US equities, mining stock-level factor data for standalone strategies and for a factor pool feeding machine learning pipelines
- Developed multivariate time-series models across Chinese and US equity index universes, with predictive modeling, statistical analysis, optimization, and data quality calibration
- Designed and co-developed the quantitative research database and strategy backtesting environment used across the quantitative team

Pricing and Analytics (Shenzhen) Limited

Shenzhen, China

Quantitative Analyst: Derivatives Pricing and Valuation

2017.12–2019.10

- Designed pricing and valuation models for mortgage-backed products including REITs and CLOs; both models were registered as software copyrights
- Applied Black-Scholes, risk-neutral valuation, and martingale pricing methods, and implemented extreme value theory and Value at Risk for portfolio tail risk

TECHNICAL SKILLS

Programming: Python (PyTorch, pandas, NumPy), R, C++, SQL

Econometrics: time-series and realized volatility modeling on tick-level data, forecast evaluation and comparison, quantile methods, long-memory and fractional integration

Machine learning: deep sequence models (attention, convolutional, recurrent, set-based); out-of-sample validation and model selection

Quantitative finance: portfolio construction and optimization, dependence modeling with copulas, tail risk (extreme value theory, VaR, CVaR), derivatives pricing and valuation

Computing: GPU cluster workloads; reproducible large-scale experiment pipelines

Languages: English (proficient), Cantonese (proficient), Mandarin (native)

REFEREES

Assoc. Prof. Vitali Alexeev, Finance, University of Technology Sydney (PhD supervisor)

Assoc. Prof. Katja Ignatieva, Actuarial Studies, UNSW (PhD co-supervisor)

Assoc. Prof. Zhenyu Cui, Financial Engineering, Stevens Institute of Technology (master's project supervisor)